Midterm 2 W26

2026-03-12

Instructions

This is an in-person exam. Your exam must be complete and uploaded to Gradescope by 2:00p.

Answer the following questions and complete the exercises in RMarkdown. Please embed all of your code and push your final work to your repository. Your code must be organized, clean, and run free from errors. Remember, you must remove the `#` for any included code chunks to run. Be sure to add your name to the author header above.

Your code must knit in order to be considered. If you are stuck and cannot answer a question, then comment out your code and knit the document. You may use your notes, labs, and homework to help you complete this exam. Do not use any other resources- including AI assistance or other students' work.

Don't forget to answer any questions that are asked in the prompt! Each question must be coded; it cannot be answered by a sort in a spreadsheet or a written response only.

For all plots you create, a title and clearly labeled axes must be provided.

Be sure to push your completed midterm to your repository and upload the document to Gradescope. *Incorrectly uploading the exam to Gradescope will result in a 5 point deduction.* This exam is worth 50 points.

Load the libraries

```
library(tidyverse)
library(janitor)
library(naniar)
library(ggmap)
library(lubridate)
library(shiny)
```

API Key

Since you will build a map as part of the exam, take a moment to register your API key. You may need to look up your stadia maps API key online (please do that now).

```
#register_stadiamaps("824e1e2a-26b0-4511-874e-b93a502f5dce", write = FALSE)
```

About the data

This study is focused on the fasting behavior of polar bears in the Southern Beaufort (SB) and Chukchi seas (CS). The data were collected from 1983-2017 and include measurements of blood urea nitrogen (BUN) and creatinine, which are used to calculate a ratio that indicates whether a bear is fasting or not. The original paper is included in the data folder. *It may help to quickly read the abstract before starting the exam.*

The data are from: Rode, K.D., Wilson, R.R., Douglas, D.C., Muhlenbruch, V., Atwood, T.C., Regehr, E.V., Richardson, E.S., Pilfold, N.W., Derocher, A.E., Durner, G.M., Stirling, I., Amstrup, S.C., St. Martin, M., Pagano, A.M., and others, 2017, Spring fasting behavior in a marine apex predator provides an index of ecosystem productivity: Global Change Biology, v. 24, no. 1, p. 410–423, https://doi.org/10.1111/gcb.13933.

Data resources

- https://data.usgs.gov/datacatalog/data/USGS:ASC141
- https://www.sciencebase.gov/catalog/item/61a1a9e7d34eb622f69743d2

The variables include:

- `bear_id`: A unique number given to each bear.
- `capture_date`: The data when the bear was tranquilized and assessed.
- `popn`: The population to which the bear belongs.
- `sex`: Sex of each bear.
- `age_class`: The age class of each bear (A=adult, S=subadult).
- `mating`: Whether or not the bear is pursuing a mate.
- `denning`: Whether or not the bear denned last year.
- `lat`: Latitude where the bear was located.
- `long`: Longitude where the bear was located.
- `urea_nitrogen`: Concentration of urea nitrogen.
- `creatinine`: Concentration of creatinine.

Load the data

Run the code chunk below to load the data, clean the names, and do some tidying.

```
polar_bears <-
  read_csv("data/polarBear_serumUreaCreatinine_beaufortChukchi_rode.csv") %>%
  clean_names() %>%
  select(-den_method, -no_cubs, -cub_age) %>%
  filter(age_class==c("A", "S")) %>%
  mutate(sex=toupper(sex),
         age_class=toupper(age_class))
```

1. (2 points) List the variable names and use a summary function to get an idea of the data structure.

```r
names(polar_bears)
```

```
##  [1] "bear_id"       "capture_date"  "popn"          "sex"
##  [5] "age_class"     "mating"        "denning"       "lat"
##  [9] "long"          "urea_nitrogen" "creatinine"
```

```r
#polar_bears
```

```r
summary(polar_bears)
```

```
##     bear_id       capture_date            popn               sex
##  Min.   :  697   Min.   :1983-04-23   Length:519         Length:519
##  1st Qu.:20167   1st Qu.:2000-04-10   Class :character   Class :character
##  Median :20705   Median :2008-04-12   Mode  :character   Mode  :character
##  Mean   :18040   Mean   :2004-07-02
##  3rd Qu.:21220   3rd Qu.:2011-03-31
##  Max.   :99990   Max.   :2016-04-22
##   age_class            mating            denning               lat
##  Length:519         Length:519         Length:519         Min.   :62.46
##  Class :character   Class :character   Class :character   1st Qu.:68.11
##  Mode  :character   Mode  :character   Mode  :character   Median :70.53
##                                                           Mean   :69.86
##                                                           3rd Qu.:71.00
##                                                           Max.   :72.70
##       long         urea_nitrogen     creatinine
##  Min.   :-172.2   Min.   : 1.00   Min.   :0.350
##  1st Qu.:-165.2   1st Qu.: 6.00   1st Qu.:0.850
##  Median :-150.7   Median :11.00   Median :1.050
##  Mean   :-153.7   Mean   :15.13   Mean   :1.079
##  3rd Qu.:-146.5   3rd Qu.:21.25   3rd Qu.:1.285
##  Max.   :-134.8   Max.   :69.00   Max.   :2.350
```

```r
#glimpse(polar_bears)
```

2. (3 points) How are NA's represented and where are they distributed in the variables? Use the `naniar` package to visualize the distribution of NA's in the data and replace any alternative representations of NA with actual NA values.

```r
#NA's are represented as "unknown"

miss_var_summary(polar_bears)
```

```
## # A tibble: 11 x 3
##    variable      n_miss pct_miss
##    <chr>          <int>    <num>
##  1 bear_id            0        0
##  2 capture_date       0        0
##  3 popn               0        0
##  4 sex                0        0
##  5 age_class          0        0
```

```
## 6 mating              0        0
## 7 denning             0        0
## 8 lat                 0        0
## 9 long                0        0
## 10 urea_nitrogen      0        0
## 11 creatinine         0        0
```

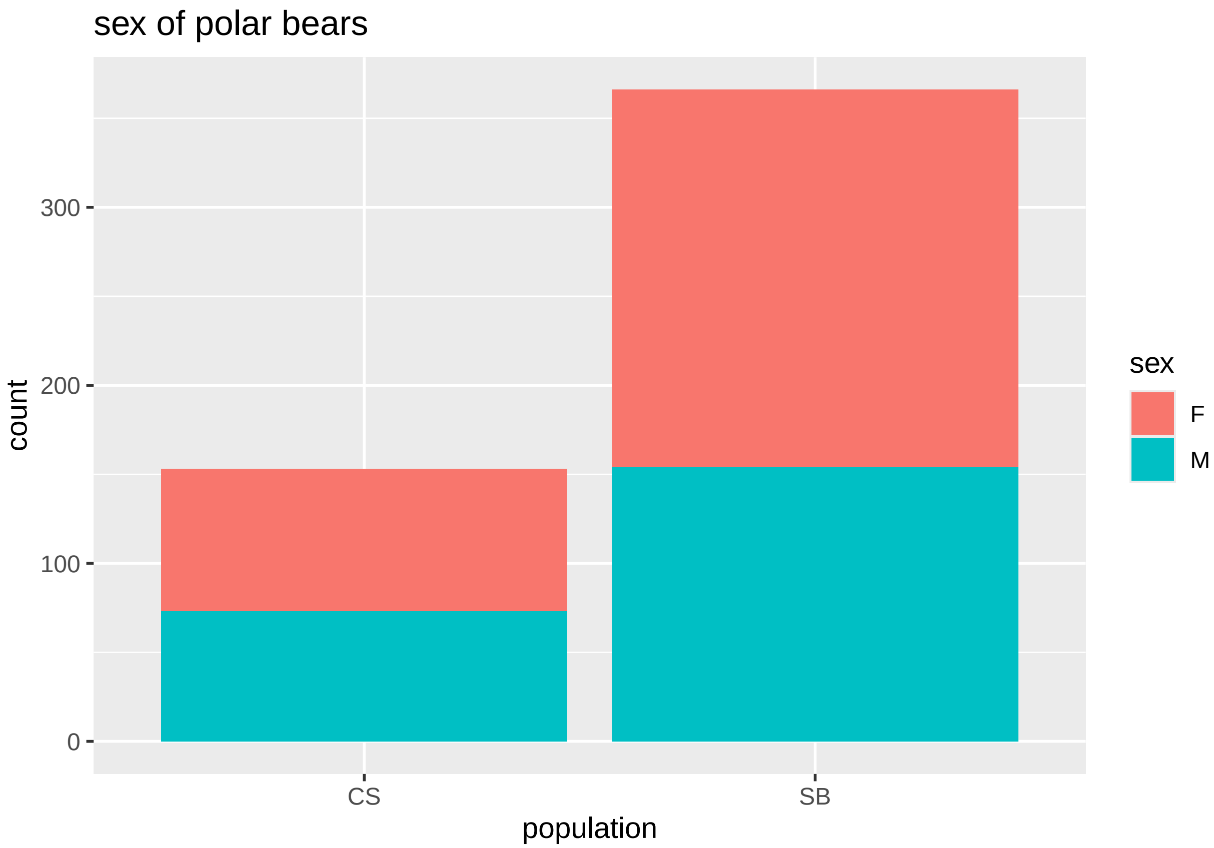
```
ppol_bears<- polar_bears %>%
  mutate(mating=na_if(mating, "unknown"))
```

```
pol_bears<- polar_bears %>%
  mutate(denning=na_if(denning, "unknown"))
```

```
polar_bears_new<- c(pol_bears, ppol_bears)
```

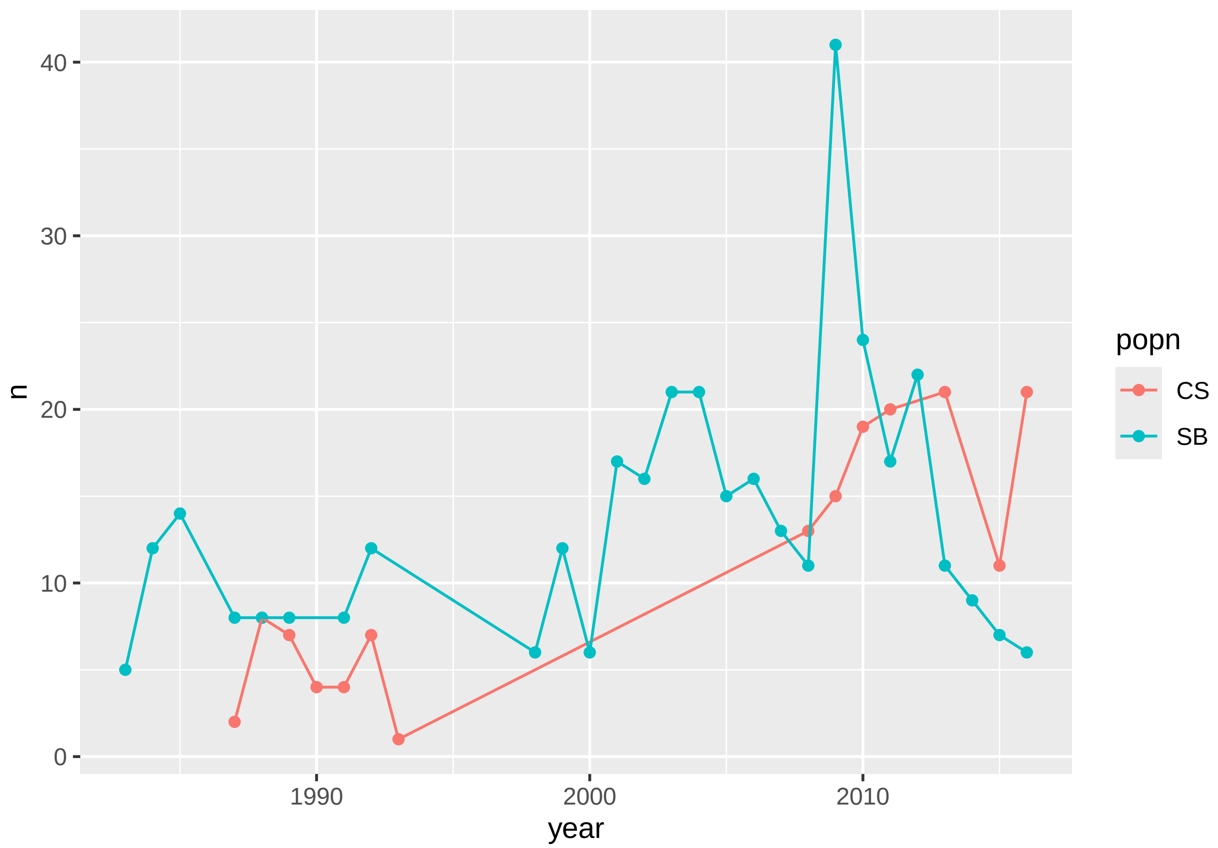
3. (6 points) Is sampling even for both sexes? Make a count the shows the number of males and females sampled by population and make a plot to visualize the counts.

```
#There are more females than males
polar_bears %>%
  group_by(sex, popn)%>%
  summarise(count=n(), .groups = "drop_last")%>%
  ggplot(aes(x = popn, y = count, fill = sex, position = "dodge"))+
  geom_col()+
  labs( title = "sex of polar bears",
        x = "population",
        y = "count")
```



4. (6 points) How many bears were sampled in each year for both populations? Make a line plot(s) to visualize sampling by year for both populations. (hint: you will need to pull apart the date to get the year in its own column)

```
#split date column then group by year
  polar_bears%>%
  mutate(date = ymd(capture_date))%>%
  separate(capture_date, into = c("year", "month", "day"), sep = "-")%>%
    mutate(year = as.numeric(year)) %>%
    count(year, popn)%>%
  ggplot(aes(x= year, y= n, color = popn))+
  geom_point()+
  geom_line()
```




```
polar_bears<- polar_bears%>%
  mutate(date = ymd(capture_date))%>%
  separate(capture_date, into = c("year", "month", "day"), sep = "-")%>%
    mutate(year = as.numeric(year))
```

5. (2 points) What are the min, median, mean, and max values for `urea_nitrogen` and `creatinine`?

```
polar_bears %>%
  group_by(urea_nitrogen) %>%
  summary(mean_urea=mean(urea_nitrogen, na.rm=T),
          median_urea=median(urea_nitrogen, na.rm=T),
          min_urea=sd(urea_nitrogen, na.rm=T),
          max_urea=max(urea_nitrogen,na.rm = T),
          n=n())
```

```
##     bear_id           year          month               day
##  Min.   :  697   Min.   :1983   Length:519         Length:519
##  1st Qu.:20167   1st Qu.:2000   Class :character   Class :character
##  Median :20705   Median :2008   Mode  :character   Mode  :character
##  Mean   :18040   Mean   :2004
##  3rd Qu.:21220   3rd Qu.:2011
##  Max.   :99990   Max.   :2016
##      popn               sex              age_class            mating
##  Length:519         Length:519         Length:519         Length:519
##  Class :character   Class :character   Class :character   Class :character
```

```
##  Mode  :character   Mode  :character   Mode  :character   Mode  :character
##
##
##
##    denning              lat              long         urea_nitrogen
##  Length:519         Min.   :62.46   Min.   :-172.2   Min.   : 1.00
##  Class :character   1st Qu.:68.11   1st Qu.:-165.2   1st Qu.: 6.00
##  Mode  :character   Median :70.53   Median :-150.7   Median :11.00
##                     Mean   :69.86   Mean   :-153.7   Mean   :15.13
##                     3rd Qu.:71.00   3rd Qu.:-146.5   3rd Qu.:21.25
##                     Max.   :72.70   Max.   :-134.8   Max.   :69.00
##    creatinine          date
##  Min.   :0.350   Min.   :1983-04-23
##  1st Qu.:0.850   1st Qu.:2000-04-10
##  Median :1.050   Median :2008-04-12
##  Mean   :1.079   Mean   :2004-07-02
##  3rd Qu.:1.285   3rd Qu.:2011-03-31
##  Max.   :2.350   Max.   :2016-04-22
```

```
#Min= 1.00, Max=69.00, Median= 11.00
```

```
  polar_bears %>%
  group_by( creatinine) %>%
  summarize(mean_cre=mean(creatinine, na.rm=T),
            median_cre=median(creatinine, na.rm=T),
            min_cre=min(creatinine, na.rm=T),
            max_cre=max(creatinine,na.rm = T),
            n=n())
```

```
## # A tibble: 53 x 6
##    creatinine mean_cre median_cre min_cre max_cre     n
##         <dbl>    <dbl>      <dbl>   <dbl>   <dbl> <int>
##  1       0.35     0.35       0.35    0.35    0.35     1
##  2       0.4      0.4        0.4     0.4     0.4      5
##  3       0.45     0.45       0.45    0.45    0.45     3
##  4       0.47     0.47       0.47    0.47    0.47     2
##  5       0.5      0.5        0.5     0.5     0.5      5
##  6       0.55     0.55       0.55    0.55    0.55     8
##  7       0.57     0.57       0.57    0.57    0.57     3
##  8       0.6      0.6        0.6     0.6     0.6      6
##  9       0.63     0.63       0.63    0.63    0.63     1
## 10       0.65     0.65       0.65    0.65    0.65    13
## # i 43 more rows
```

```
#Min= 0.350, Max= 2.350, Median= 1.050
```

6. (3 points) As explained above, the ratio of `urea_nitrogen` to `creatinine` is used as an index of fasting. To calculate the ratio, we need to multiply `urea_nitrogen` by 0.466, then divide by `creatinine`. Create a new variable called `un_creat_ratio` that contains this calculation.

```
#calculate the creat ratio for each entry, not the mean
#summary is not needed
```

```r
un_creat_ratio<- polar_bears%>%
 mutate(un_creat_ratio = (urea_nitrogen * 0.466)/creatinine)

head(un_creat_ratio)
```

```
## # A tibble: 6 x 15
##   bear_id  year month day   popn  sex   age_class mating denning   lat  long
##     <dbl> <dbl> <chr> <chr> <chr> <chr> <chr>     <chr>  <chr>   <dbl> <dbl>
## 1    6398  1985 04    28    SB    F     A         no     yes      70.4 -148.
## 2    6384  1991 04    13    SB    F     A         no     yes      70.0 -150.
## 3    6520  1992 04    18    SB    F     A         no     yes      70.9 -145.
## 4    6559  2000 04    12    SB    F     A         no     yes      70.0 -142.
## 5   20629  2003 04    16    SB    F     A         no     yes      70.5 -145.
## 6   20876  2007 04    01    SB    F     A         no     yes      71.1 -153.
## # i 4 more variables: urea_nitrogen <dbl>, creatinine <dbl>, date <date>,
## #   un_creat_ratio <dbl>
```

7. (4 points) The authors of the study categorized bears as fasting if their `un_creat_ratio` was less than or equal to 10. Use `case_when()` to create a new variable called `fasting` that categorizes bears as fasting or not.

```r
#create a new column and name it fasting
#also make a case where a bear is not fasting
 un_creat_ratio %>%
  mutate(fasting = case_when(un_creat_ratio <= 10 ~ "fasting", un_creat_ratio > 10 ~ "not fasting")) %>%
  mutate(as.factor(fasting))
```

```
## # A tibble: 519 x 17
##    bear_id  year month day   popn  sex   age_class mating denning   lat  long
##      <dbl> <dbl> <chr> <chr> <chr> <chr> <chr>     <chr>  <chr>   <dbl> <dbl>
##  1    6398  1985 04    28    SB    F     A         no     yes      70.4 -148.
##  2    6384  1991 04    13    SB    F     A         no     yes      70.0 -150.
##  3    6520  1992 04    18    SB    F     A         no     yes      70.9 -145.
##  4    6559  2000 04    12    SB    F     A         no     yes      70.0 -142.
##  5   20629  2003 04    16    SB    F     A         no     yes      70.5 -145.
##  6   20876  2007 04    01    SB    F     A         no     yes      71.1 -153.
##  7   20354  2009 04    16    SB    F     A         no     yes      70.7 -145.
##  8   21024  2009 04    25    SB    F     A         no     yes      70.5 -147.
##  9   20764  2009 05    01    SB    F     A         no     yes      70.8 -149.
## 10   21045  2009 05    19    SB    F     A         no     yes      70.7 -148.
## # i 509 more rows
## # i 6 more variables: urea_nitrogen <dbl>, creatinine <dbl>, date <date>,
## #   un_creat_ratio <dbl>, fasting <chr>, `as.factor(fasting)` <fct>
```

```r
#head(fasting)
```

```r
#true_fasting <- fasting
```

8. (6 points) The authors found that the number of fasting bears increased in the SB population but decreased in the CS population. Make a plot(s) that shows the number of fasting bears by year for both populations to visualize this trend.

```
#un_creat_ratio%>%
#  filter(fasting == "fasting")%>%
#  ggplot(aes(x = year, y = popn))+
#  geom_point()
```

9. (8 points) Where are the polar bears used in this study located? Make a static (not leaflet) distribution map that shows the locations of bears by population. (hint: use `color=popn` in the `aes()` of `geom_point()`)

```
#polar_bears %>%
#  select(lat, long) %>%
#summary()
```

```
#lat <- c(62.46, 72.70)
#long <- c(-172.2, -134.8)
#bbox <- make_bbox(long, lat, f=0.05)
```

```
#map1<- get_stadiamap(bbox, maptype = "stamen_terrain", zoom = 5)

#ggmap(map1)
```

```
#ggmap(map1)+
#  geom_point(data = polar_bears, aes(long, lat), size = 0.5, color = "black", alpha= 0.8)+
#  labs(x= "longitude",
#       y= "latitude",
#       title = "polar bears")
```

10. (10 points) Build a simple shiny app that shows the range of `un_creat_ratio` values for both populations. The user should be able to select the fill variable and choose between sex, mating, age_class, denning, and fasting. Please use the code chunk below so that your exam knits.

```
#library(shiny)

#ui <- fluidPage(

#    dashboardHeader(title = "population of polar bears"),

#  dashboardSidebar(disable = TRUE),

#  dashboardBody(
#  selectInput("x",
#              "select fill variable",
#              choices = c("sex",
 #                         "mating",
#                          "age_class",
#                          "denning",
 #                         "fasting"),
#              selected = "sex"),
#    plotOutput("plot",
#             width = "500px",
#             height = "400px")
```

```
)

#server <- function(input, output, session) {
#  output$plot <- renderPlot({

 #output$plot<- renderPlot({

 #     ggplot(polar_bears,
#       aes_string(x = input$x, y= un_creat_ratio, fill = "un_creat_ratio"))+
 #   geom_boxplot()
}
)}
#shinyApp(ui, server)
```

Submit the Midterm

1. Save your work and knit the .rmd file.
2. Open the .html file and "print" it to a .pdf file in Google Chrome (not Safari).
3. Go to the class Canvas page and open Gradescope.
4. Submit your .pdf file to the midterm assignment- be sure to assign the pages to the correct questions.
5. Commit and push your work to your repository.